

Auteur: Luca Letroye
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$$\int \frac{x-1}{\sqrt{x+2}} dx$$

$$t = x + 2$$

$$\frac{dt}{dx} = 1$$

$$= \int \frac{t-3}{\sqrt{t}} dt$$

$$= \int t^{\frac{1}{2}} dt - 3 \int t^{\frac{-1}{2}} dt$$

$$= \frac{2t^{\frac{3}{2}}}{3} - 6t^{\frac{1}{2}} + C$$

$$= \frac{2(x+2)^{\frac{3}{2}}}{3} - 6(x+2)^{\frac{1}{2}} + C$$

References